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SUPERSEDING
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22 July 1970

MILITARY SPECIFICATION

MAGNESIUM POWDER (FOR USE IN AMMUNITION)

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers powdered magnesium suitable for use in ammunition. (see 6.1)

1.2 Classification. Magnesium powder shall be furnished in the following types, grades, and granulations:

1.2.1 Type.

Type I - Flaked and/or chip
Grade A (96 percent minimum magnesium content)
Grade B (86 percent minimum magnesium content)

Type II - Oblong chip with rounded edges
(96 percent minimum magnesium content)

Type III - Atomized
(98 percent minimum magnesium content)

Type IV - Flaked and/or chip (See 6.5)
(95.5 percent minimum magnesium content)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U'S Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement

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1.2.2 Granulation.

Granulation No.	Nominal mesh size
1	40 to 80
2	40 to 80 (alternate)
3	50 to 100
4	50 to 100 (Army)
5	50 to 120
6	80 to 120
7	100
8	120 to 200
9	140
10	200
11	80 to 200
12	120 to 200 (Army)
13	20 to 50
14	50 to 100
15	100 to 200
16	200 to 325
17	50 to 100
18	30 to 50

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Standard for Testing Purposes

MILITARY

MIL-C-6054 - Container: Steel, Shipping

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MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-1218 - ACS Chemicals

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 100-199 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite, "49 CFR 100-199).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
B-329 - Apparent Density of Refractory Metals and Compounds by the Scott Volumeter
D1193-Reagent Water
E-300 - Sampling Industrial Chemicals

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street Philadelphia, Pennsylvania 19103).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I (grade A), Types II and III. Type I (grade A), type II and III magnesium powder shall be manufactured from primary magnesium metal containing not less than 99.80 percent metallic magnesium. Mica fillers or other adulterants shall not be used and "bag house dust" shall not be included in the magnesium powder offered for acceptance.

3.1.2 Type I (grade B). Type I (grade B) magnesium powder shall be manufactured from magnesium metal of such purity that the product meets the requirements of this specification. Mica fillers or other adulterants shall not be used and "bag house dust" shall not be included in the magnesium powder offered for acceptance.

3.1.3 Type IV - Type IV magnesium powder shall be manufactured from primary magnesium metal containing not less than 95.50 percent metallic magnesium (See 6.5). Mica fillers or other adulterants shall not be used and "bag house dust" shall not be included in the magnesium offered for acceptance.

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3.2 Form.

3.2.1 Type I. Type I magnesium powder shall consist of shavings, turnings, flakes, or any combination of these which meet the necessary granulation requirements shown in Table II.

3.2.2 Type II. Type II magnesium powder shall consist of oblong chip-like particles with rounded edges.

3.2.3 Type III. Type III magnesium powder shall consist of granular or spheroidal particles.

3.2.4 Type IV. Type IV magnesium powder shall consist of shavings, turnings, flakes, or any combination of these which meet the granulation 7 requirements given in Table II.

3.3 Chemical composition. Magnesium powder shall conform to the requirements shown in Table I when determined in accordance with methods given in 4.5

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Table I. Chemical composition

	Type I Grade A	Type I Grade B	Type II	Type III	Type IV
	Percent	Percent	Percent	Percent	Percent
Volatile at 105°C (max)	0.10	0.10	0.10	0.10	0.10
Oil and grease (max)	0.02	0.02	0.02	0.02	0.02
Carbides (max)	0.004	0.004	0.004	0.004	0.004
Material insoluble in sulfuric acid (max)	0.15	-	0.15	0.15	1.00
Metallic iron (max)	0.05	-	0.05	0.05	0.05
Total iron as Fe_2O_3 (max)	0.25	-	0.25	0.25	0.25
Aluminum as Al (max)	-	10	-	-	2.50
Free metallic magnesium (min)	96	-	96	98	95.50
Aluminum plus free metallic magnesium (min)	-	96	-	-	-

3.4 Granulation. Magnesium powder shall be of the fineness shown in Table II when determined as specified in 4.5.11. All percentages shall be by weight using US Standard Sieves conforming to RR-S-366. The powder shall pass through the required sieves readily without balling or the particles clinging together.

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Table I. Granulation

U.S. Standard Sieve No.		Granulation 1	Granulation 2	Granulation 3	Granulation 4	Granulation 5	Granulation 6	Granulation 7	Granulation 8	Granulation 9	Granulation 10	Granulation 11	Granulation 12	Granulation 13	Granulation 14	Granulation 15	Granulation 16	Granulation 17	Granulation 18
Through	Retained	Percent																	
		Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.	Max.	Min.
6	8																		
8	12																		
12	20											5	99						
16	30											50	80						
20	50				100							80	50						
25	70				75														
30	100		100				80		100										
40	140	10	7					1											
50	200	35	30				100												
70	270	10	15				10												
80	325		5				90												
100	400								15										
120	450																		
140	500																		
170	625																		
200	750																		
230	900																		
270	1000																		
325																			

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(see p. 1 and p. 4)

3.5 Apparent density. The apparent density of the magnesium powder when determined as specified in 4.5.12 shall be as shown in Table III.

Table III. Apparent density

Granulation No.	Density in grams per milliliter	
	Maximum	Minimum
1	0.65	0.55
2	0.65	0.55
3	0.75	0.65
4	0.625	0.45
5*	-	-
6*	-	-
7*	-	-
8*	-	-
9*	-	-
10*	-	-
11	-	0.45
12	-	0.45
13	-	0.70
14	0.75	0.65
15	-	0.90
16	-	0.90
17	-	0.90
18*	-	-

* No determination required.

3.6 Analysis of metal. A chemical analysis of the metal used in the manufacture of magnesium powder shall be furnished with each lot offered for acceptance. (See 6.2 and 6.5)

3.7 Workmanship.

3.7.1 Process.

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3.7.1.1 Types I, II, and IV. - Type I, II and IV magnesium powder shall be manufactured by grinding, stamping, turning, or any combination of these methods which will yield a suitable product.

3.7.1.2 Type III. - Type III magnesium powder shall be manufactured by the atomizing process.

3.7.2 Care shall be exercised to avoid inclusion of foreign material during manufacture.

3.8 First Article Inspection. - This specification contains provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspection. - The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3)
b. Quality Conformance Inspection (see 4.4)

4.3 First Article Inspection

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of 10 kg of the magnesium powder obtained by sampling as described in 4.4.2.1. The samples shall be obtained from a production lot which has been produced by the contractor using the same producion processes, procedures and equipment as will be used in fulfilling the

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contract. All materials, shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be Performed. - The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.5 of this specification.

4.3.3 Rejection. - See MIL-A-48078

4.4 Quality Conformance Inspection

4.4.1 Lot Formation. - A lot shall consist of one or more batches of the magnesium powder produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of magnesium powder that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105. In the event the process is a batch operation, each batch shall be a lot.

4.4.2 Testing.

4.4.2.1 Sampling.

4.4.2.1.1 Sampling of Shipping Containers (see 5.) - Sampling of shipping containers shall be conducted in accordance with MIL-STD-105.

4.4.2.1.2 Sampling of Magnesium Powder. - Two (2) independent representative samples, approximately 2 kg, shall be obtained from each lot of magnesium powder using ASTM Procedure E300. When lots comprise 3 or less containers, all containers in the lot shall be selected. Before sampling, the contents of each container (selected for sampling) shall be thoroughly blended by rolling and turning end for end. If either of these two samples fail to meet any of the test requirements, the lot shall be rejected. Any remaining samples

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shall be held for a minimum of one (1) year for possible future examination. The samples shall be subjected to the tests listed below.

TABLE IV

Inspection/Requirement	Defect Classification
Form	Major
Chemical Composition	Major
Granulation	Major
Apparent Density	Major

4.4.3 Inspection Equipment. - For the performance of all tests and examinations specified in 4.5, commercial inspection equipment should be employed. The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes. (see 6.3)

4.5 Test Methods and Procedures for Quality Conformance (see 6.4).

4.5.1 General Notes.

4.5.1.1 Test and Examinations. - Unless otherwise specified herein, all chemicals shall be ACS grade in accordance with MIL-STD-1218. Also, the distilled water used shall be in accordance with ASTM-D-1193.

4.5.1.2 Test Methods and Procedures. - (see the tests in 4.5.2 thru 4.5.12). - Test methods and procedures shall be performed, using prescribed analytical procedures for replicate determinations given in standard analytical textbooks.

4.5.1.3 PRECAUTION. - This specification covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

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the specific issue of individual documents referenced (see 2.1

4.5.2 Form. - Place a small portion of the sample on a glass slide and examine under a 20 to 30 power microscope.

4.5.3 Material volatile at 105°C. Accurately weigh into a tared dish approximately 5.0g. of the sample, heat for 3 hours at 105°C., cool and weigh. Calculate the percent material volatile at 105°C as follows:

$$\text{Percent material volatile at } 105^{\circ}\text{C} = \frac{100A}{W}$$

where: A = loss in weight, g.
W = weight of sample, g.

4.5.4 Oil and grease. - Extract a 50.0 g. sample with 250 ml. of chemically pure ethyl ether (redistilled over sodium) for 1 hour in a Soxhlet extractor or similar apparatus having an extracting tube of 250 ml. capacity. Evaporate on steam bath to a volume of approximately 50 ml. and filter the extract into a tared 150 ml. beaker. Rinse the flask with three 10 ml. portions of ethyl ether. Make a blank determination for residue of the either. Cover the beaker with a ribbed watch glass. Evaporate the ether on a water bath that is constantly at boiling temperature. The total time of heating on the water bath should be approximately 1 hour. Remove from the water bath and dry the outside of the beaker with a clean cloth. Heat in an oven at 100° to 105°C for 30 minutes, cool and weigh. Calculate the percent oil and grease as follows:

$$\text{Percent oil and grease} = \frac{100\ (A-B)}{W}$$

where: A = weight of residue, g.
B = weight of blank residue, g.
W = weight of sample, g.

Note. Considerable care shall be exercised through the entire procedure to prevent contamination of the sample or the extract with grease or dust from the hands or other source.

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4.5.5 Carbides.

4.5.5.1 Apparatus. Use an apparatus similar to that shown in Figure 1.

4.5.5.2 Hydroxylamine hydrochloride reagent. - Dissolve 0.25 g. of gelatin in hot water. Dilute to 500 ml. and add 500 ml. of 95 percent ethyl alcohol and 1.25 gm. hydroxylamine hydrochloride.

4.5.5.3 Procedure. Add an accurately weighed portion of approximately 50 g. of magnesium powder to the reaction flask (A). Transfer 50 mg. of cuprous chloride to test tube (B). Pass nitrogen through the apparatus to remove the air and continue the passage of nitrogen throughout the determination. Lower the test tube and add 15 ml. of concentrated NH_4OH and 30 ml. of hydroxylamine hydrochloride reagent. Raise the test tube, insert stopper, add 100 ml. of water from dropping funnel (C) and switch on the heater. Heat the mixture in the reaction flask just to boiling. A red or pink coloration of the cuprous ammonium chloride solution in the test tube indicates that the magnesium sample contains more than 0.004 percent carbides. Should the cuprous ammonium chloride solution become blue, due to the introduction of air, it must be decolorized with successive 5 ml. portions of hydroxylamine hydrochloride reagent until the blue color disappears.

4.5.6 Material insoluble in sulfuric acid.- Accurately weigh and transfer approximately 5.0g of the sample to a 600 ml. beaker and dissolve in 150 ml of 1:5 H_2SO_4. Allow to stand until action ceases and all of the magnesium has dissolved. Filter through a tared crucible and wash thoroughly with small portions of hot water, being careful to keep the

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volume of filtrate and washings at approximately 200 ml. Reserve the filtrate for determination of total iron (see 4.5.8). Heat the crucible at 105°C for 2 hours, cool and weigh. Calculate the percent material insoluble in sulfuric acid as follows:

$$\text{Percent material insoluble in sulfuric acid} = \frac{100A}{B}$$

where: A = weight of residue, g
B = weight of sample, g

4.5.7 Metallic Iron.

4.5.7.1 Wet Chemical Method. - Using 100 g. of sample, pass a magnet or an electromagnet through the powder until no more particles are taken up. Brush the particles adhering to the magnet onto a sheet of glazed paper and separate the magnetic from the nonmagnetic by passing the magnet under the paper and drawing the magnetic particles aside. Repeat the procedure to eliminate as much as possible any adhering particles of magnesium. Transfer the magnetic particles onto a balance pan, weigh, and report as metalic iron. In the event that ths weight exceeds the requirements for the grade of powder being tested, analyze for iron as follows: Transfer the weighed material to a 250 ml. beaker and add 100 ml. of 5 percent sulfuric acid. Boil until dissolved. Cool. Pass the solution through a Jones reductor, taking the usual precautions to activate the amalgamated zinc. Wash with 50 ml. of 5 percent sulfuric acid and 150 ml. of water. Titrate with standard 0.02N potassium permangnate to a pink color. Calculate the percent metallic iron as follows:

$$\text{Percent metallic iron} = \frac{5.58\ VN}{W}$$

where: V = Volume of $KMnO_4$ used, ml.
N = Normality of $KMnO_4$ solution.
W = Weight of sample, g.

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4.5.7.2 Atomic Absorption (Alternate Method). - Determine the metallic iron content of the magnesium powder by an Atomic Absorption method which has prior approval of the Contracting Officer. (see 6.4)

4.5.8 Total iron. Transfer the filtrate from the determination of insoluble matter (see 4.5.6) to a 300 ml. Erlenmeyer flask, add a spiral of aluminum wire, cover with a small watch glass and boil for approximately 10 minutes. Cool under running water, remove the aluminum spiral and titrate immediately with N/20 KMnO4 solution to a pink color. Run a blank determination on the aluminum using 150 ml. of 1:5 H_2SO_4. Calculate the percent total iron as follows:

$$\text{Percent total iron (as } Fe_2O_3) = \frac{7.98\ (V-B)N}{W}$$

where: V = Volume of $KMnO_4$ used for sample, ml.
B = Volume of $KMnO_4$ used for blank, ml.
N - Normality of KMnO4 solution.
W = Weight of sample, g.

4.5.9 Aluminum.

4.5.9.1 Wet Chemical Method. - Transfer an accurately weighed portion of approximately 0.40g of the magnesium powder sample to a 400 ml. beaker. Cover with a watch glass and add 10 ml. of concentrated H_2SO_4 down the lip of the beaker. Add 50 ml. of water very slowly, taking care to prevent the reaction from proceeding too rapidly. When the reaction has subsided, boil to dissolve the last particles of metal. If there is any residue, add HNO_3 to clear. Filter the solution if necessary, wash and dilute to about 250 ml. Add 25 ml. of a saturated solution of NH_4Cl and a few drops of methyl red indicator. Heat nearly to boiling and add 1:4 NH_4OH until the color just changes to yellow. Boil for about 1 minute and immediately filter through a Whatman No 40 or equivalent filter paper. Wash twice with hot 2 percent NH_4Cl solution. Place the paper

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and precipitate back into the original beaker. Add 20 ml. of HCl (1:1), digest on steam bath for a few minutes, and break up the filter paper with a stirring rod. Dilute to about 200 ml. with water and precipitate as before. Police and wash several times with hot 2 percent NH_4Cl solution. Transfer the precipitate to a tared platinum crucible. Burn the filter paper at dull red heat and ignite at 1000° to 1100°C for 1 hour. Cool in a desiccator and weigh as rapidly as possible. Calculate the percent aluminum as follows:

$$\text{Percent aluminum} = \frac{52.93A}{W}$$

where: A = Weight of residue, g.
W = Weight of sample, g.

4.5.9.2 Atomic Absorption (alternate Method). - Determine the percent aluminum of the magnesium powder by an Atomic Absorption method which has prior approval of the Contracting Officer (see 6.4).

4.5.10 Free metallic magnesium.

4.5.10.1 Gas Absorption Method.

4.5.10.1.1 Apparatus.

(a) 1 graduated dropping funnel.
(b) 1 generating flask.
(c) 1 20 inch reflux condenser.
(d) 1 glass-stoppered gas washing bottle containing concentrated H_2SO_4.
(e) 1 drying column containing dehydrite or equivalent.
(f) 1 combustion furnace having the tube projected from the furnace approximately 6 inches on the inlet side and 2 and 1/2 to 3 inches on the outlet side. The tube shall be filled with copper oxide held in place by a small asbestos plug.

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(g) 1 absorption tube containing dehydrite or equivalent.
(h) 2 guard bottles containing concentrated H_2SO_4.
(i) 1 capsule for weighing and introducing the sample. (This may be conveniently made by cutting off a small vial 3/4 inch from the bottom.
(j) 1 suction line.

4.5.10.1.2 Assembly of apparatus. Connect the gas generating flask to the bottom of the dropping funnel using a Y tube having one side open as an air inlet. Connect the lower end of the reflux condenser to the generating flask and the upper end to the gas washing bottle. Connect the lower end of the drying column to the gas washing bottle and the upper end to the inlet of the combustion furnace. Connect the lower end of the absorption tube to the outlet of the combustion furnace and the upper end to the two guard bottles connected in series. Connect the outlet of the second guard bottle to the suction line.

4.5.10.1.3 Procedure. Plug in the furnace, adjust the rheostats to heat the furnace quickly to 600°C and turn on the water in the reflux condenser. In approximately 1 hour turn on the suction to draw air through the train at the rate of approximately 5 bubbles per second in the gas washing bottle. Weigh the absorption tube to obtain the initial weight, reconnect it into the train, draw air through it for 45 minutes, disconnect, cool for 30 minutes, and reweigh. A blank of 0.4 mg. or less is considered satisfactory. Transfer 1.000 gm of the sample into the capsule and carefully slide it into the generating flask, being careful not to scatter the powder. Reconnect the absorption tube, adjust the rheostats to maintain the furnace at 580° to 620°C and carefully add approximately 60 ml of distilled water, enough to cover the lower end of the inlet tube, through the dropping funnel. Place 30 ml. of 30 percent H_2SO_4 in the dropping funnel and begin the decomposition of the sample slowly by adding the H_2SO_4 a drop or two at a time, being careful to note that the bubbles are constantly entering the inlet tube

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at the rate of approximately 5 bubbles per second. It is advisable to have on hand a large beaker filled with cold water in which the generating flask may be immersed if the reaction becomes too violent. After 10 to 15 ml. of H_2SO_4 have been added, the reaction is nearly complete and the remainder of the acid may be added rapidly. Gently boil the contents of the flask for 3 minutes, draw air through for an additional 30 minutes, disconnect the absorption tube, cool, and weigh. Disconnect the generating flask and rinse the dropping funnel and the inlet tube well with distilled water to remove acid. Calculate the percent metallic magnesium as follows:

$$\text{Percent free metallic magnesium} = \frac{135\ (A-B)}{W}$$

where A = gain in weight of absorption tube, g.
B = blank, g.
W = weight of sample, g.

4.5.M6.2 Alternate procedure for determination of free metallic magnesium (eudiometer method). When desired, the following eudiometer method may be used: Assemble an apparatus similar to that shown on figure 2. The gas measuring buret shall be graduated in 0.2 ml. The reaction flask shall contain 100 ml. of 3 percent H_2SO_4 which has been saturated with hydrogen gas. Means shall be provided for dropping the sample into the acid. The weight of the sample to be used will depend on the metal content and the temperature and pressure at which hydrogen is measured. At standard conditions of temperature and pressure, a 0.47 g sample of pure magnesium will yield about 470 ml. of hydrogen. The weight of sample should be roughly computed to give a volume of hyrdogen within the capacity of the graduated portion of the gas buret. The system must be in equilibrium with respect to temperature and pressure and checked for any gas leakage before the determination is started. The weighed sample is folded in a piece of paper and placed in position. The temperature is recorded after comparing the reading of the

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thermometer in the reaction flask with a thermometer located in the close vicinity of the apparatus. The volume is also recorded after the pressure is equalized by means of the leveling bulb. The confining liquid in the pipet is water saturated with hydrogen gas and containing 0.1 percent H_2SO_4 and a few drops of methyl orange to facilitate the observation of the solution in the apparatus. Allow the sample to drop into the acid and shake. After the reaction has ceased, again bring the system to equilibrium by cooling all parts of system to the initial temperature ($\pm$ 1°C) and equalizing the pressures. Record the volume of hydrogen gas evolved. Calculate to percent free metallic magnesium as follows:

Percent free metallic magnesium =

$$\frac{\frac{(P-Pw)}{760} \times \frac{273}{(273+T)} \times 0.001086 \times 100}{W} = \frac{0.03901A(P-Pw)}{W(273+T)}$$

where: A = reading of gas buret in ml.
P = reading of barometer in mm. Hg.
Pw = vapor pressure of water at temperature T
T = temperature of the system at equilibrium in degrees Centigrade
W = g of sample

4.5.10.3 Atomic Absorption Method (Alternate Method). - Determine the following metallic ions present in the magnesium powder sample by an Atomic Absorption method which has prior approval of the Contracting Officer (see 6.4)

Aluminum
Manganese
Zinc
Silicon
Beryllium
Copper
Nickel

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Sum the values obtained above and use this sum as A in the calculation given below:

4.5.10.3.1 Calculation. - Calculate the percent free metalic magnesium (FMM) as follows:

$$\%FMM = 100\% - [\%A + \%B + \%C + \%D + \%E + \%F]$$

where: A = % Metallic Ion Impurities (see 4.5.10.3)
B = % Volatiles (see 4.5.3)
C = % Oil and Grease (see 4.5.4)
D = % Carbides (see 4.5.5)
E = Material Insoluble in Sulfuric Acid (see 4.5.6)
F = % Total Iron as Fe_2O_3 (see 4.5.8)

4.5.11 Granulation. - Place a weighed portion of approximately 50 g. of the sample on the top sieve of a nest of sieves assembled as specified in Table II and provided with a bottom pan. Cover and shake for 30 minutes in a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the portions retained by each sieve and calculate to a percentage basis as required.

4.5.12 Apparent Density. - Determine the apparent density of the magnesium powder in accordance with the method described in ASTM B329

5. PREPARATION FOR DELIVERY

5.1 Application. - The packing and marking requirements specified herein apply only to direct purchase by or direct shipment to the government.

5.2 Packing. - Unless otherwise specified, level A packing shall be employed.

5.2.1 Level A. - Unless otherwise specified in the contract or order, magnesium powder shall be packed in unlined containers conforming to MIL-C-6054, class I, IA, II or III and the requirements of Code of Federal Regualtions 49 CFR 100-199.

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5.2.2 Level B. - Not applicable.

5.2.3 Level C. - The packing shall be adequate to afford protection against damage during direct shipment from supply source to the first receiving activity for immmediate use. The containers shall comply with the Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation, and the Code of Federal Regulations 49 CFR 100-199.

5.3 Marking. - In addition to any other special marking required shipments shall be marked in accordance with MIL-STD-129 and should include the lot number.

6. NOTES

6.1 Intended use. - Magnesium powder covered by this specification is intended for use in the manufacture of tracer, igniter, signal flare and pyrotechnic compositions. Type II material is required for the manufacture of bursting igniters. In general, higher density material will give longer burning times in tracers.

6.2 Ordering data. - In addition to the requirements given in MIL-A-48078 (PA), the following are required:

a. Type, grade and granulation
b. A Certificate of Analysis of the magnesium powder in accordance with MIL-STD-1171
c. A Certificate of Analysis of metal used (see 3.6)

6.3 Submission of Inspection Equipment Designs for approval See MIL-A-48078. - Submit equipment designs, as required, to Commander, US Army Armament Research and Development Command, ATT: DRDAR-QA, Dover, New Jersey 07801.

6.4 Equivalent Test Methods - Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the Contracting Officer to:

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Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QA, Dover, New Jersey 07801. This description should include but not be limited to the procedures used, the accuracy and precision and drawings of any special equipment required.

6.5 Type IV - The composition for the alloy used in the manufacture of type IV magnesium powder should be as follows:

Magnesium, %	95.50 min
Aluminum, %	2.0 - 2.50
Manganese, %	0.30 - 0.50
Zinc, %	0.15 - 0.21
Silicon, %	0.60 - 1.20
Beryllium, %	0.0005 0 0.0015
Copper, %	0.04 max
Nickel, %	0.01 max
Other, %	0.30 max

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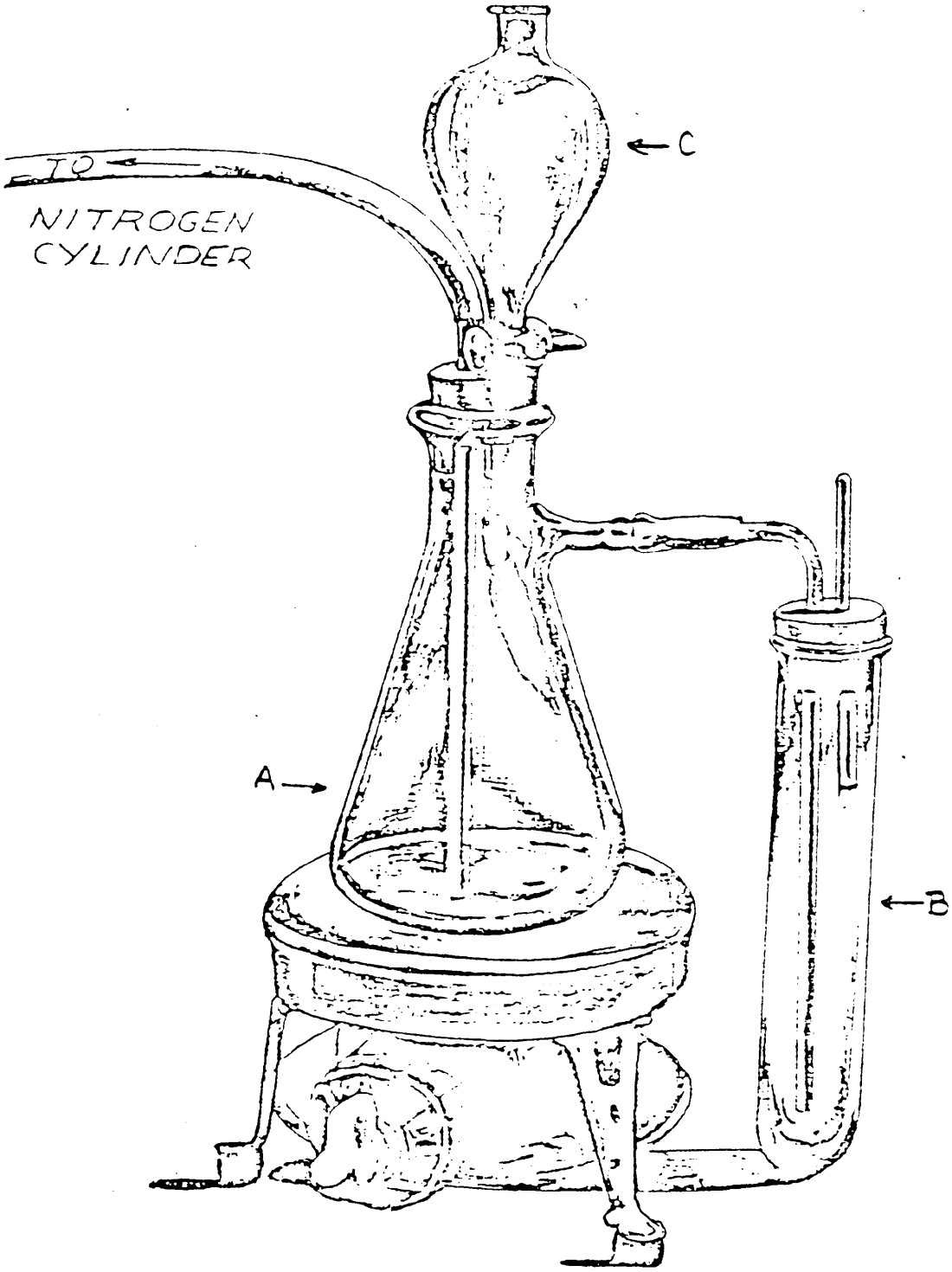


Figure 1. Apparatus for Determination of Carbides in Magnesium

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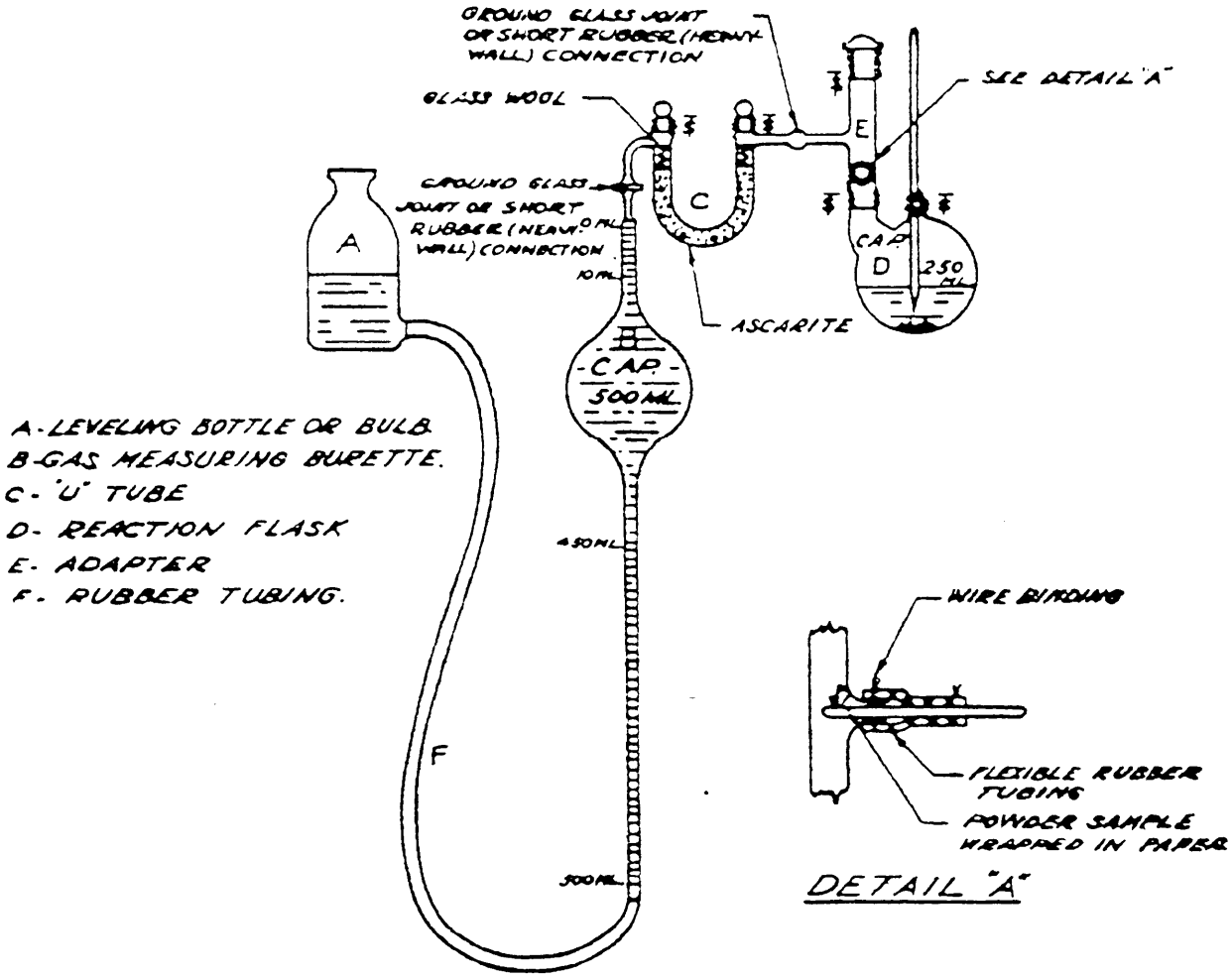


FIGURE 2.—Eudiometer for Determination of Magnesium Metal in Magnesium Powder.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-M-382C Magnesium Powder

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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and physical characteristics of the product in conformance with